Programming Exam 3

Date/Time: 2026.03.31 08:00 – 08:50

(程式檔命名**學號_quiz3.py**，上傳至 Moodle PE3 上傳區)

Problem: SpaceX Rocket Fueling Cost Estimator

Create a Python program called **SpaceX Rocket Fueling Cost Estimator**. The program helps a SpaceX launch operator estimate how much fuel is needed and how much it will cost to prepare a rocket for launch. To keep the problem simple, use the following fictional simplified model for the rocket and fueling system. Your program should repeatedly allow the user to estimate fueling costs until the user decides to stop.

When the program starts, it should greet the user. Then it should ask the user to enter the rocket's **current fuel percentage** and **target fuel percentage**. After that, the user chooses a fueling mode. Based on the chosen mode, the program calculates:

- The amount of fuel needed
- The estimated fueling cost

Finally, the program asks whether the user wants to perform another estimation.

<u>Requirements</u>

(1) **Welcome Message**

When the program starts, display a welcome message such as:

"Welcome to the SpaceX Rocket Fueling Cost Estimator!"

(2) **Ask for current and target fuel percentages**

Prompt the user to enter:

- The current fuel percentage of the rocket
- The target fuel percentage of the rocket

Both values should be stored as floating-point numbers.

(3) **Check valid percentage range**

The current fuel percentage and target fuel percentage must both be between 0 and 100. If either value is outside this range, display an error message and ask the user to enter the values again.

(4) **Check fueling logic**

The target fuel percentage must be greater than the current fuel percentage. If not, display an error message and ask the user to enter the values again.

(5) **Ask the user to choose a fueling mode**

The program should let the user choose one of the following fueling modes:

- 1 = Standard Ground Fueling
- 2 = Rapid Launch Fueling
- 3 = Cryogenic Precision Fueling

(6) **Use the following fixed values**

Assume the rocket has a total fuel capacity of **1200 tons**.

Each fueling mode has a different cost per ton and efficiency:

- Standard Ground Fueling
  Cost: $18 per ton
  Efficiency: 95%
- Rapid Launch Fueling
  Cost: $25 per ton
  Efficiency: 90%
- Cryogenic Precision Fueling
  Cost: $30 per ton
  Efficiency: 92%

(7) Calculate required fuel using this formula

required_fuel = (target_percent - current_percent) / 100 * 1200 / efficiency

(8) **Calculate total fueling cost** using this formula

total_cost = required_fuel * cost_per_ton

(9) **Display the result. Show:**

- The selected fueling mode
- The estimated fuel required
- The estimated fueling cost

Display fuel with **2 decimal places**, and cost with **2 decimal places**.

(10) **Display an extra message in one case**

If the user selects **Rapid Launch Fueling** and the required fuel is more than **500 tons**, display this message: "***Warning: Large rapid fueling operation detected.***"

(11) **Ask whether the user wants another estimation**

After displaying the result, ask:

"***Would you like to estimate another fueling session? (yes/no):***"

(12) **Repeat or end**

- If the user enters **"yes"**, repeat the process.
- If the user enters **"no"**, display a goodbye message and end the program.

Sample Input/Output (以下是你程式執行後須印出的結果)

```
c:\workspace>python pe3.py
Welcome to the SpaceX Rocket Fueling Cost Estimator!

Enter current fuel percentage (0-100): 20
Enter target fuel percentage (0-100): 80

Select a fueling mode:
1. Standard Ground Fueling
2. Rapid Launch Fueling
3. Cryogenic Precision Fueling
Enter your choice (1-3): 1

--- Fueling Summary ---
Fueling Mode: Standard Ground Fueling
Estimated Fuel Required: 757.89 tons
Estimated Fueling Cost: $13642.11

Would you like to estimate another fueling session? (yes/no): no
Thank you for using the SpaceX Rocket Fueling Cost Estimator!

c:\workspace>
```

```
c:\workspace>python pe3.py
Welcome to the SpaceX Rocket Fueling Cost Estimator!

Enter current fuel percentage (0-100): 40
Enter target fuel percentage (0-100): 95

Select a fueling mode:
1. Standard Ground Fueling
2. Rapid Launch Fueling
3. Cryogenic Precision Fueling
Enter your choice (1-3): 2

--- Fueling Summary ---
Fueling Mode: Rapid Launch Fueling
Estimated Fuel Required: 733.33 tons
Estimated Fueling Cost: $18333.33
Warning: Large rapid fueling operation detected.

Would you like to estimate another fueling session? (yes/no): no
Thank you for using the SpaceX Rocket Fueling Cost Estimator!

c:\workspace>
```

```
c:\workspace>python pe3.py
Welcome to the SpaceX Rocket Fueling Cost Estimator!

Enter current fuel percentage (0-100): 90
Enter target fuel percentage (0-100): 60
Target percentage must be greater than current percentage.

Enter current fuel percentage (0-100): 30
Enter target fuel percentage (0-100): 85

Select a fueling mode:
1. Standard Ground Fueling
2. Rapid Launch Fueling
3. Cryogenic Precision Fueling
Enter your choice (1-3): 3

--- Fueling Summary ---
Fueling Mode: Cryogenic Precision Fueling
Estimated Fuel Required: 717.39 tons
Estimated Fueling Cost: $21521.74

Would you like to estimate another fueling session? (yes/no): yes

Enter current fuel percentage (0-100): 10
Enter target fuel percentage (0-100): 50

Select a fueling mode:
1. Standard Ground Fueling
2. Rapid Launch Fueling
3. Cryogenic Precision Fueling
Enter your choice (1-3): 1

--- Fueling Summary ---
Fueling Mode: Standard Ground Fueling
Estimated Fuel Required: 505.26 tons
Estimated Fueling Cost: $9094.74

Would you like to estimate another fueling session? (yes/no): no
Thank you for using the SpaceX Rocket Fueling Cost Estimator!

c:\workspace>
```

(繳交是交 pe3.py 檔，不是交截圖)

Note: You need to write comments (註解) for each part in your code.